



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Textile Technology
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: TY B. Tech.

Course: Man Made Fibre Technology

Course Code: PCC-TT305

Date: 19.08.2025

Time: 15:00 to 15:45

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	Figure 1 Shows the schematic diagram of melt spinning machine. Figure 1			
a	A spinneret with 100 holes is used to produce filaments at a polymer flow rate of 0.5 g/min per hole. Calculate the total throughput rate.	2	CO1	BT1
b	Discuss various factors influence the spin ability of a polymer.	2	CO1	BT2
Q.2	High-speed melt spinning involves the rapid solidification and stretching of molten polymer filaments, leading to changes in fibre structure.			
a	Discuss the steps involved in the development of fibre structure during high-speed melt spinning with the help of suitable diagram.	6	CO2	BT3
b	Discuss the effects of neck-line deformation and localized orientation on fibre structure formation in high-speed melt spinning.	4	CO2	BT3
Q.3	A textile company is planning to spin a new polymer variant of PET (Polyethylene terephthalate) that softens at 240°C and degrades at 280°C. As a process engineer answer the following questions.			

a	Suggest a most suitable synthesis route for this PET variant, emphasizing environmental sustainability and low chemical emissions. Justify it	4	CO3	BT2
b	Analyse the impact of Di-ethylene Glycol (DG) formation during the polymerization process on the processing and performance of the PET variant.	2	CO3	BT3

==== ALL THE BEST ====

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Determine suitable method and parameters for spinning of a filament
CO 2	Determine process temperature looking to thermal behavior of a fiber
CO 3	Compare properties of various fibers for probable application
CO 4	Select suitable spin finish for a fiber and process
CO 5	Select high performance fiber for special applications
CO6	Determine thermal effect on fiber properties and structure



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Textile Technology
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: TY B. Tech.	Course: Man Made Fibres	Course Code: PCC-TT303
Date: 19.08.2025	Time: 15:00 to 15:45	Max. Marks: 20

Note:

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- v. The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	A textile company is planning to spin a new polymer variant of PET (Polyethylene terephthalate) that softens at 240°C and degrades at 280°C. As a process engineer answer the following questions.				
	a	Suggest a most suitable synthesis route for this PET variant, emphasizing environmental sustainability and low chemical emissions. Justify it.	4	CO1	BT1
	b	Analyse the impact of Di-ethylene Glycol (DG) formation during the polymerization process on the processing and performance of the PET variant.	2	CO1	BT2
Q.2	High-speed melt spinning involves the rapid solidification and stretching of molten polymer filaments, leading to changes in fibre structure.				
	a	Discuss the steps involved in the development of fibre structure during high-speed melt spinning with the help of suitable diagram.	6	CO2	BT3
	b	Discuss the effects of neck-line deformation and localized orientation on fibre structure formation in high-speed melt spinning.	4	CO2	BT3
Q.3	Figure 1 Shows the schematic diagram of melt spinning machine. Figure 1				

a	A spinneret with 100 holes is used to produce filaments at a polymer flow rate of 0.5 g/min per hole. Calculate the total throughput rate.	2	CO3	BT2
b	Discuss various factors influence the spin ability of a polymer.	2	CO3	BT3

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Determine suitable method and parameters for spinning of a filament
CO 2	Determine process temperature looking to thermal behavior of a fiber
CO 3	Compare properties of various fibers for probable application
CO 4	Select suitable spin finish for a fiber and process
CO 5	Select high performance fiber for special applications
CO6	Determine thermal effect on fiber properties and structure



S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded.

In-Semester Examination-I AY 2025-26

SGGSJET, Nanded

Date: 18/08/2025

Class : Third Year Textile Technology

Time: 15.00 PM

Max. Marks : 20

Subject: PCC-TT303 Fabric Forming Technology- III

Duration: 45 Min

- Note: i. Attempt any 4 questions from below questions.
ii. Assume suitable data if necessary

Q.N.	Question	Marks	CO	BT
1	List and classify the various weaving systems.	5	1	1
2	Discuss the different types of rapier looms and their advantages and disadvantages.	5	4	2
3	Explain the Gabler type of weft insertion system with a schematic diagram.	5	2	2
4	Explain the semi-positive let-off motion.	5	2	2
5	Explain the pirn changing mechanism in cimmco automatic loom with neat sketch	5	2	2
6	Explain the disadvantages associated with shuttle looms and how these are overcome in shuttleless looms.	5	4	3

== ALL THE BEST ==

Bloom's Taxonomy (BT):

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

- CO1: Understand conventional and new weaving technologies.
CO2: Demonstrate the different principle motions of automatic and modern weaving systems
CO3: Apply knowledge of advance weaving systems in product manufacturing etc.
CO4: Compare the weaving technologies for appropriate selection.
CO5: Evaluate the techno-economics of weaving systems.

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S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded.

In-Semester Examination-I AY 2025-26

Date: 18/08/2025

Class : Third Year Textile Technology

Time: 15.00 PM

Max. Marks : 20

Subject: PCC-TT302 Fabric Forming Technology- III

Duration: 45 Min

- Note: i. Attempt any 4 questions from below questions.
ii. Assume suitable data if necessary

Q.N.	Question	Marks	CO	BT
1	List and classify the various weaving systems.	5	1	1
2	Discuss the different types of rapier looms and their advantages and disadvantages.	5	4	2
3	Explain the Gabler type of weft insertion system with a schematic diagram.	5	2	2
4	Explain the semi-positive let-off motion.	5	2	2
5	Explain the pirn changing mechanism in cimmco automatic loom with neat sketch	5	2	2
6	Explain the disadvantages associated with shuttle looms and how these are overcome in shuttleless looms.	5	4	3

== ALL THE BEST ==

Bloom's Taxonomy (BT):

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

- CO 1: Understand conventional and new weaving technologies.
CO 2: Write about features, principle motions of automatic and modern weaving systems
CO 3: Apply knowledge of advance weaving systems in product manufacturing etc.
CO 4: Compare the weaving technologies for appropriate selection.

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Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of Textile Technology
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: TY

Course: Textile Testing-I

Course Code: PCC-TT301

Date: 20.08.2025

Time: 15:00 to 15:45

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
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- Figures to right indicate full marks.
- The symbols carry usual meaning.
- All questions carry equal marks. Q. 3 & Q.4 are compulsory questions. Attempt any one question between Q.1 & Q.2 and Q.5 & Q.6.

Q.N.		Marks	CO	BT
Q.1	Derive the relationship between yarn count and yarn diameter.	5	CO1	BT5
OR				
Q.2	Derive the relationship between yarn count and twist multiplier.	5	CO1	BT5
Q.3	Discuss in details the effect of twist on properties of yarn & fabric.	5	CO3	BT2
Q.4	a Define random and biased sampling.	2	CO1	BT2
	b Explain the cut square method of fibre sampling.	3	CO1	BT3
Q.5	One hundred and fifty pieces of a certain yarn were tested for strength, the mean value of the test results being 60 N and their standard deviation was 3 N. Calculate the 95 % confidence limits for the 'true' mean strength of the yarn. If it was required to estimate the 'true' mean correct to 0.5, how many tests should be performed?	5	CO4	BT5
OR				
Q.6	To estimate the mean count of a delivery of yarn, five standard count tests were carried out, with the following results (tex). Calculate the 99 % confidence limits for the mean count. if it was required to estimate the mean count to within 0.18 tex, how many tests should be carried out ? Tex Values: 41.3, 41.1, 40.9, 41.2, and 41.3	5	CO4	BT5

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand the principle and method of measurement of various physical and mechanical properties of fibre, yarn and fabrics in the field of textiles.
CO 2	Apply the knowledge of these measurement techniques for future applications.
CO 3	Analyse physical and mechanical properties of fibre, yarn and fabrics in the field of textiles.
CO 4	Conduct experiments related various measurement and performance evaluation of the textile materials.
CO 5	Propose and design innovative testing methods or modification for improved evaluation of textile materials in specific applications.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Textile Technology
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: TY

Course: Spun Yarn Technology-III

Course Code: PEC-TT301

Date: 22.08.2025

Time: 15:00 to 15:45

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.
- All questions carry equal marks. Attempt any four questions.

Q.N.		Marks	CO	BT
Q.1	Explain the concept of back doubling in rotor spinning.	5	CO1	BT2
Q.2	Explain the effect of rotor groove and doffing tube on the properties of rotor spun yarn.	5	CO2	BT4
Q.3	Discuss in details the structure of rotor yarn and its influence on yarn properties.	5	CO2	BT4
Q.4	Discuss the raw material requirements for air-jet spinning.	5	CO1	BT2
Q.5	Compare the structure and properties of air-jet yarns with ring spun yarns.	5	CO2	BT4

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand fundamentals of new spinning technologies and processes such as Rotor, Air-jet and Fiction spinning along with other minor technologies.
CO 2	Analyze various process and machine parameters and their influence on respective output materials.
CO 3	Evaluate the process and conduct experiment related to the process and its performances.
CO 4	Design and manufacture different types of spun and engineered yarns from apparel to technical and industrial applications using various technologies.
CO 5	Compare and contrast advanced spinning systems such as self-twist, compact spinning, and fancy yarn production methods, understanding their unique properties and applications.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Textile Technology
Summer Term Examination
Academic Year: 2024-25

Class: TY

Course: Spun Yarn Technology-II

Course Code: PCC-TT301

Date: 16.06.2025

Time: 10:00 to 13:00

Max. Marks: 100

Note:

- i. Assume suitable data whenever necessary and state it clearly.
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- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. All questions are compulsory.

Q.N.			Marks	CO	BT		
Q.1		Write a short note on	10	CO1	BT1		
	a	Berkolisation					
	b	Noil					
	c	Shore Hardness					
	d	False Draft					
Q.2							
	a	Why fibres must be kept parallel to each other in lap sheet?	05	CO3	BT2		
	b	A lap has a count of 50 Ktex and that 10% of its mass is extracted as a noil, i. Find new linear density of lap after the noil extraction. ii. Find draft due to noil extraction.	05	CO4	BT4		
Q.3	a	What are the objectives of draw frame? Explain.	10	CO3	BT2		
Q.4	a	Explain close loop auto leveller with neat labelled diagram.	10	CO2	BT2		
Q.5	a	Explain electro-mechanical building mechanism in speed frame.	10	CO2	BT3		
Q.6	a	How lap thickness affect the combing process?	10	CO3	BT3		
Q.7	a	Find surface speeds, main draft and break draft of a draw frame machine. If, rpm & dia. of rollers are as follows:	10	CO4	BT4		
		1 st roller				953 rpm	35 mm
		2 nd roller				1334 rpm	25 mm
		3 rd roller				347 rpm	20 mm
		4 th roller				267 rpm	20 mm
		5 th roller				201 rpm	28 mm
Q.8	a	If the spindle of a speed frame machine runs at 1400 rpm, the front bottom roller has a diameter of 28 mm and runs at a speed of 250 rpm, the bare bobbin dia. is 1.96 inch & full bobbin dia. is 5.90 inch. Find the rpm of bare bobbin & full bobbin.	15	CO4	BT3		
Q.9	a	Explain combing cycle with neat labelled diagram	15	CO5	BT2		

=== ALL THE BEST ===

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